

The MFU AP-MAP analysis system

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ABSTRACT

This computer engineering project focuses on analyzing Mae Fah Luang University's wireless connectivity system, with a primary goal of identifying blind spots crucial for network administrators. Ensuring complete coverage and user satisfaction within the wireless network is imperative. The system utilizes specialized software to collect and analyze data, effectively pinpointing areas with inadequate or absent signals recognized as blind spots.

Developed by the School of Information Technology, this system offers a valuable tool empowering network administrators to optimize wireless coverage and enhance user experiences significantly.

1. INTRODUCTION

Mae Fah Luang University (MFU) is a large campus with many buildings, and wireless connectivity is essential for students, faculty, and staff to access online resources and communicate with each other. However, due to the complex nature of wireless signals, there may be areas on the campus where wireless signals are weak or non-existent, also known as dead spots.

Analyzing wireless signals to identify dead spots is an important task for network administrators to ensure that wireless coverage is adequate and meets users' needs. Therefore, MFU's School of Information Technology has developed a system to analyze wireless signals and identify dead spots on the campus. The system works by using software to collect data and analyze it to identify areas with weak or non-existent signals, which will be marked as a dead point.

In summary, analyzing wireless signals to identify dead spots is an important task in ensuring a reliable wireless network on the MFU campus. The system developed by the School of Information Technology

provides a valuable tool for network administrators to optimize wireless coverage and improve the user.

2. LITERATURE REVIEW

a. Coverage and Capacity Planning Theory

Coverage and Capacity Planning Theory serves as a foundational framework for network planners and administrators to strategically deploy access points, ensuring comprehensive coverage, adequate capacity, and optimal network performance based on thorough analysis and estimations of coverage area, user density, and traffic demands.

b. Signal Propagation Models

Signal Propagation Models are instrumental in predicting signal behavior, allowing network planners to anticipate areas with potential signal weaknesses. These models help determine optimal access point placement and configurations, ensuring adequate signal coverage and strength for reliable wireless communication within a network.

c. Spatial Analysis And GIS

Spatial Analysis and GIS Techniques offer valuable insights into the geographical aspects of a network. By visualizing and analyzing spatial data, network planners gain a better understanding of user distributions and coverage patterns, aiding in strategic decisions regarding access point deployment, optimizing network coverage, and ensuring efficient wireless communication across different geographic areas within the network.

d. Traffic Engineering and Network Optimization

Traffic Engineering and Network Optimization theories help in understanding and managing traffic demands within a network infrastructure. By estimating traffic patterns, optimizing resource allocation, and ensuring efficient traffic flow, these theories contribute to effective network planning, especially in determining the

number and placement of access points to handle expected traffic demands effectively.

e. Machine Learning and Predictive Analytics

Machine Learning and Predictive Analytics provide valuable insights into future network demands and user behavior. By leveraging historical data and advanced algorithms, these techniques assist in making informed decisions regarding access point deployments, optimizing network resources, and ensuring the network's ability to meet evolving user needs effectively.

These references and theories encompass various aspects of network planning, signal analysis, user behavior, and predictive modeling, providing insights into determining the number of access points needed and identifying areas where additional access points should be deployed. Consulting these resources can offer valuable insights into theoretical frameworks supporting data analysis for access point usage and deployment in networks.

3. ANALYSIS AND DESIGN

a. Existing System

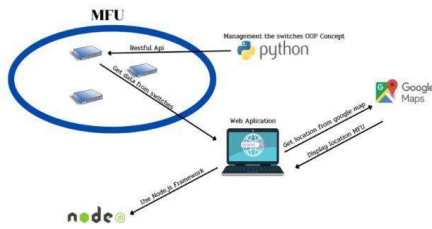


Figure 1 Existing System

With the current system, we have data available to display the status and user count of access points. However, we are unable to specify which specific access points need to be installed or deactivated at any given time to better accommodate users. This situation necessitates us to make guesses, which may lead to challenges in asset allocation and potential issues, such as slow internet speed, if there is an excessive number of users. The lack of precise information on when and where additional access points are needed or when deactivation is required poses a challenge in optimizing the system to better meet user needs. This uncertainty may result in difficulties in asset allocation and potential problems, such as slow internet speeds, when the number of users is high.

b. New System

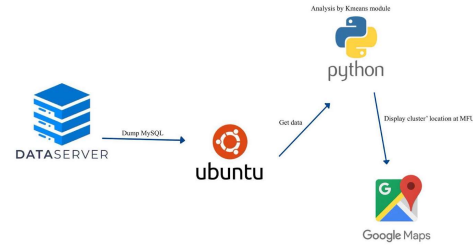


Figure 2 New System

After extracting data from the wifimap2000 network clone for storage in Ubuntu, we proceeded to conduct analysis using Python and the scikit-learn library, leveraging the KMeans module. The primary objective of this analysis is to identify locations suitable for the installation of additional access points. The results are visually presented through graphs, providing in-depth information about the latitude and longitude coordinates conducive to strategically placing new access points. Additionally, the outcomes are showcased on Google Maps, enhancing spatial context and user-friendly presentation.

c. Use case diagram

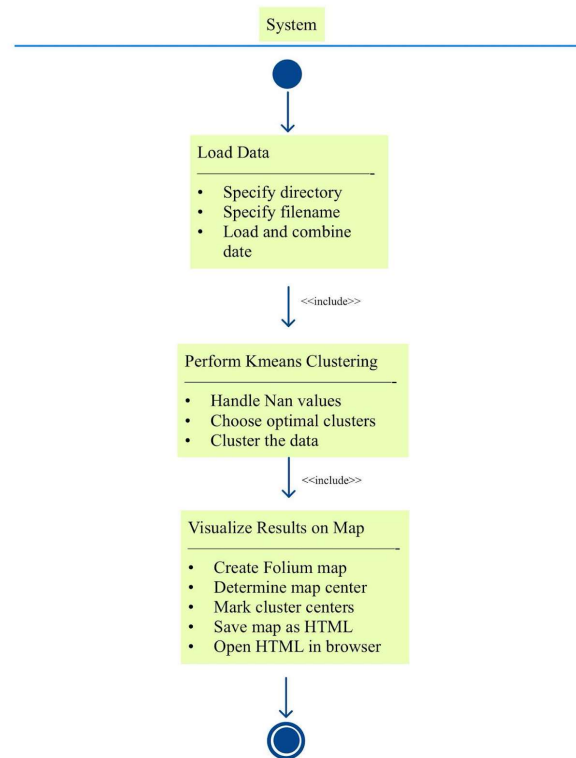


Figure 3 Use case diagram

This diagram represents the relationships between the user or external system and the three main use cases: Load Data, Perform KMeans Clustering, and Visualize Results on Map. The "<<include>>" notation signifies that the functionalities described in each subsequent use case are included within the higher-level use case.

d. Activity diagram

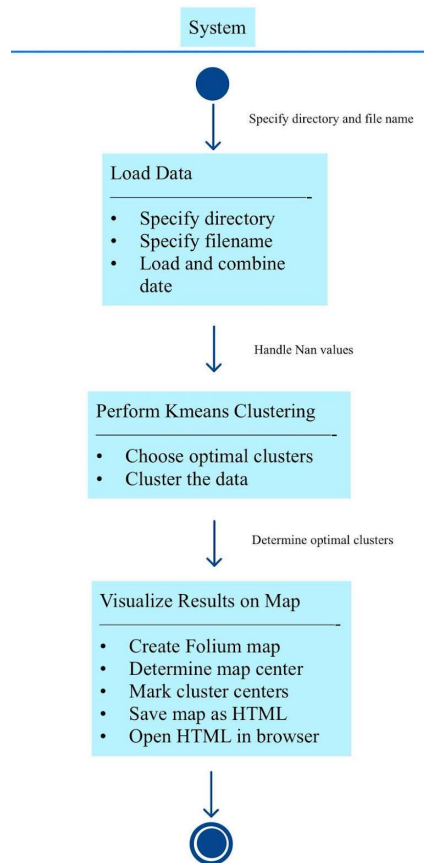


Figure 4 Activity diagram

This activity diagram representation outlines the sequence of actions within the system. The activities start from the "System" and go through the phases of "Load Data," "Perform KMeans Clustering," "Visualize Results on Map," and finally conclude with an "End" action. Each activity lists the specific tasks or operations performed as part of that phase.

4. CONCLUSION

a. Conclusion

Throughout this project, extensive research was dedicated to pivotal subjects such as signal reception, methodologies for data collection, and their practical implementation. The exploration also delved into

programming languages and software pertinent to this field, fostering a comprehensive grasp of their applicability.

A crucial focal point of this endeavor was the evaluation of the necessity for signal boosters for Mae Fah Luang University's wireless LAN controller network. This strategic plan was crafted with the primary objective of optimizing the existing network infrastructure.

The data collection process involved gathering diverse information dispersed by access points to an array of electronic devices, ranging from smartphones to tablets, laptops, and various other gadgets. Implemented within this framework was a database designed to automatically capture data upon device connection to a WiFi signal. This data was then transmitted to a centralized database for analysis.

b. Discussion

The project encountered several challenges, primarily due to its continuation from a previous senior project. One significant hurdle was understanding the code left behind by the prior team, which was essential for reconstructing the database. Additionally, aggregating user data from access points across different areas posed another obstacle. Initially, navigating various software programs posed difficulties. However, these challenges were tackled by seeking information from online sources and consulting project advisors for guidance.

Understanding the intricacies of the previous codebase and the complexities of collecting data from network broadcasters presented initial setbacks. Nonetheless, a dedicated commitment to learning, coupled with resourcefulness in seeking solutions online and leveraging the expertise of project advisors, played a vital role in overcoming these challenges. This ultimately leads to successful navigation and resolution of project complexity.

5. REFERENCES

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